

1 Co-firing copper in a cordierite-type glass-ceramic
2 printed by multi-material vat photopolymerisation: a
3 model-based process design

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5 **Abstract**

Multi-material vat photopolymerisation can print a glass-ceramic body and its copper conductors as one green part, but whether such a part can be fired is unknown. Carbon must leave the glass before its pores close, copper must stay metallic, and both materials must shrink compatibly, all in one temperature range. A verified model couples char pyrolysis and steam gasification, viscous sintering and crystallisation of the glass, diffusional sintering of copper, and co-firing mechanics in one and three dimensions. Inputs come from the literature, and the glass was calibrated against the published IBM glass-ceramic/copper process. Of the two atmospheres compared, only steam with hydrogen held at a fixed fraction of the Cu/Cu₂O equilibrium, as in the IBM process, removes the char before the glass seals. Fine copper then densifies during the burnout, well before the glass, and the free-strain mismatch reaches 15.6%. A paste and programme optimised together use an assumed retarding effect of char on copper to release both materials at once, cutting the mismatch to 3.7%, but this design is fragile. A design that tolerates a factor of two in gasification rate and 15 K in the glass transition holds the mismatch at 9–11% and the copper at 89% IACS in a 16 h cycle. In a printed system-in-package with

a die cavity, vias and an embedded solenoid, the two designs cut surface distortion from 516 to 114–266 μm ; copper plasticity keeps the cooling bow small. Gasification kinetics, glass viscosity, copper sintering viscosity and char yield should be measured first.

6 *Keywords:* Co-firing, Glass-ceramic, Copper metallisation, Vat
7 photopolymerisation, Binder removal, Sintering mismatch

8 **1. Introduction**

9 Packages for millimetre-wave front ends, antenna-in-package modules
10 and wide-bandgap power devices now have to route conductors in three
11 dimensions through a low-loss dielectric, and to do it close to the die [1,
12 2, 3, 4]. Co-fired ceramics answer part of this demand. Low-temperature
13 co-fired ceramic (LTCC) and glass-ceramic substrates combine low dielectric
14 loss, hermeticity and a thermal expansion close to that of silicon [5, 6].
15 They are, however, built from stacked tapes: conductors are printed in
16 planes, interconnections run vertically through punched vias, and every
17 change of geometry requires new tooling [7, 5]. The Heterogeneous Integration
18 Roadmap devotes a chapter to additive manufacturing for packaging, including
19 lithography-based manufacturing of ceramic parts [8].

20 The conductor of choice in fired packages has long been silver, because it
21 can be co-fired in air. Silver migrates under bias and humidity and diffuses
22 into the glass when fired in oxidising conditions [9, 10, 11]. Copper conducts
23 almost as well, costs less and is far less prone to migration, but it must be
24 fired in an atmosphere that keeps it metallic. In such an atmosphere the
25 organic binder no longer burns off cleanly and part of it is left as carbonaceous

26 residue [12]; carbon sealed in a glass that has closed its pores can bloat or even
27 burst the part [13]. IBM solved this for its glass-ceramic/copper multilayer
28 substrates with a steam-based co-sintering process that gave a dense glass-
29 ceramic with high-conductivity copper [14, 6]. That solution was developed
30 for laminated tapes.

31 Vat photopolymerisation of ceramic slurries, in particular lithography-
32 based ceramic manufacturing (LCM), produces dense ceramics with features of
33 a few tens of micrometres [15, 16, 17]. It has already been used for monolithic
34 microwave filters and antenna substrates [18, 19, 20], and photocurable LTCC
35 suspensions have been formulated for it [21]. In these parts the metallisation is
36 still added afterwards, by plating or by printing on the fired ceramic [18, 22], or
37 the conductors are jetted into ceramic layers by a separate route [23, 24, 25, 26].
38 Multi-material vat photopolymerisation removes that restriction. Printers
39 such as the CeraFab Multi 2M30 switch between slurries within a layer
40 [27, 28, 29]. Co-sintered ceramic-ceramic parts have been demonstrated [30],
41 and zirconia and titania green bodies printed by vat photopolymerisation
42 have been assembled and co-sintered in a reducing atmosphere, which made
43 the titania conductive [31]. Copper slurries for vat photopolymerisation have
44 been developed separately [32, 33, 34, 35]. In principle, then, a glass-ceramic
45 body and its copper wiring can be printed as one green part. Copper and
46 LTCC suspensions have already been inkjet-printed into a three-dimensional
47 part and co-fired [23]. For vat photopolymerisation, whose slurries carry far
48 more binder, we found no study of how such a part should be fired, and no
49 model of co-firing that includes binder removal.

50 Four requirements make that firing difficult, and they are coupled. First,

51 the char from both slurries must be removed before the glass seals its pores,
 52 in an atmosphere that does not oxidise the copper. Second, the two materials
 53 must shrink at compatible times and by compatible amounts; otherwise
 54 the part cambers, delaminates or cracks around its conductors [36, 37, 38].
 55 Third, the glass must densify before it crystallises, because crystals stop
 56 viscous flow. Fourth, a thermal-expansion mismatch of about 14 ppm K^{-1}
 57 between copper and a cordierite-type glass-ceramic must be accommodated
 58 on cooling. In tape-based LTCC the second requirement is usually met by
 59 adjusting the metal paste, for example with ceramic additions [39], or the
 60 heating schedule, guided by master sintering curves [40]. Continuum models
 61 of co-sintering reproduce camber and mismatch stresses in ceramic and metal-
 62 ceramic bilayers [41, 42, 43, 44, 45, 46], but none includes the chemistry of
 63 binder removal, and none treats copper with a crystallising glass. The first
 64 and third requirements are therefore usually handled separately from the
 65 second, although in practice they compete for the same temperature range.

66 This paper sets out a model-based design of that firing for a copper
 67 conductor printed inside a cordierite-type glass-ceramic body. We extend a
 68 slab model of binder removal and sintering, which we developed for single-
 69 material printed copper (Appendix A), to a crystallising glass-ceramic. The
 70 two materials are coupled through the viscous analogue of laminate theory
 71 and through a three-dimensional finite-element model of a voxelised printed
 72 part. The model is verified against closed-form solutions and parameterised
 73 from published data, with every input tagged by its provenance. We use
 74 it for four purposes. We (i) map the atmosphere and temperature window
 75 in which carbon can be removed from the glass before its pores close while

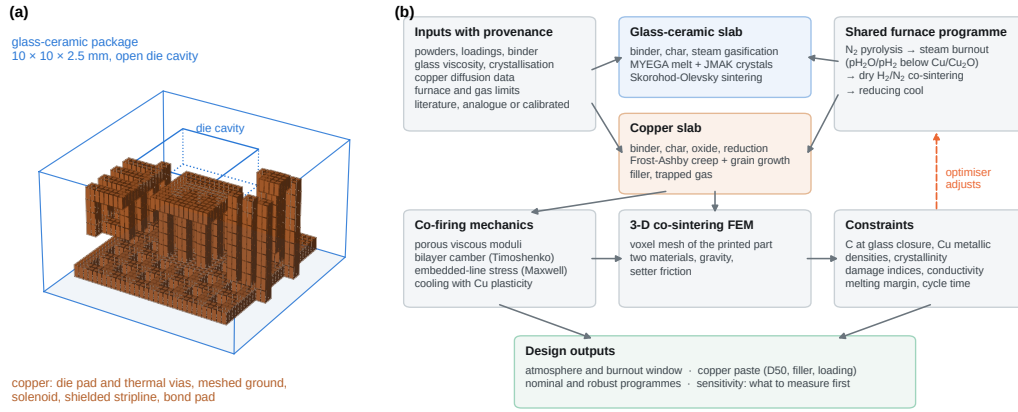


Figure 1: (a) Test vehicle, a printed radio-frequency system-in-package: a $10 \times 10 \times 2.5$ mm glass-ceramic body with an open die cavity, and copper for a die-attach pad with thermal vias, a meshed ground plane, an embedded solenoid inductor and a buried stripline with via fences and a bond pad, printed as one part (glass-ceramic shown as outline; vertical scale exaggerated). (b) Structure of the model. Each material is followed by its own slab model under a shared furnace programme; the two histories are coupled mechanically, analytically and in three dimensions, and the programme and paste are searched against constraints.

76 copper stays metallic; (ii) show that fine copper and a fragile glass densify in
 77 different temperature ranges and with different curve shapes, and quantify the
 78 consequences for camber and conductor damage; (iii) design the copper paste
 79 and the firing programme together to bring the two into step, both at nominal
 80 inputs and robustly against their uncertainty, and follow the designs through
 81 a printed test vehicle in three dimensions; and (iv) identify the few material
 82 parameters that decide the outcome, and should therefore be measured first.

2. Model formulation

2.1. Structure of the model

The co-fired part is represented at two levels (Fig. 1b). At the first level, each material is followed through the firing programme by a one-dimensional slab model that resolves heat transfer, binder decomposition, char formation and removal, pore closure and densification through the thickness of its critical section. The glass-ceramic body and the copper conductor are simulated separately but share the furnace set point and the inlet gas, so both see the same thermal and chemical history. At the second level, the two densification histories are coupled mechanically: analytically, through the viscous analogue of laminate theory, and numerically, through a three-dimensional finite-element model in which both materials occupy the voxels of the printed part. Both levels use the same constitutive law for the porous body, so a free sample in the finite-element model reproduces the slab model exactly.

The copper slab model was developed and verified for single-material copper parts printed from a lithography-based slurry; its chemistry and its calibration against published copper data are summarised in Appendix A. The additions made here are the crystallising glass-ceramic, the copper paste with an inert filler, the steam burnout atmosphere, and the co-firing mechanics.

2.2. Binder removal and residual carbon

The photocured binder is lumped into three pseudo-components i (a non-reactive solvent, a crosslinked network and a char-forming backbone), each decomposing by first-order kinetics,

$$\frac{dB_i}{dt} = -k_i(T) B_i, \quad k_i(T) = \frac{E_i \beta_{\text{ref}}}{RT_{\text{p},i}^2} \exp \left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{p},i}} \right) \right], \quad (1)$$

106 where $T_{p,i}$ is the temperature of the corresponding peak in a thermogravimetric
 107 trace at $\beta_{\text{ref}} = 10 \text{ K min}^{-1}$. Writing the rate constant in terms of the peak
 108 temperature keeps the two fitted quantities nearly independent and lets them
 109 be read directly from a single thermogravimetric curve. A fraction χ of the
 110 network and backbone mass remains as char. In a steam-bearing atmosphere
 111 the char is gasified by $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$ at the rate

$$\begin{aligned}
 r_g = & a(\varepsilon_o) (1 - f_{\text{cl}}) k_g(T) \frac{c}{M_C} \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{ref}}} \right)^{1/2} \\
 & \times \frac{1 + K_H x_{\text{H}_2}^{\text{ref}}}{1 + K_H x_{\text{H}_2}} \left(1 - \frac{Q}{K_{\text{st}}(T)} \right)_+,
 \end{aligned} \tag{2}$$

112 in which c is the char concentration, $a(\varepsilon_o) = \varepsilon_o/(\varepsilon_o + 0.02)$ the fraction of
 113 pore surface reachable through open porosity ε_o , f_{cl} the closed-pore fraction,
 114 K_H an inhibition constant for hydrogen adsorption, $Q = x_{\text{CO}}x_{\text{H}_2}/x_{\text{H}_2\text{O}}$ the
 115 reaction quotient and K_{st} the equilibrium constant from NIST-JANAF data
 116 [47]. Char in closed pores is not gasified: once the pores seal, whatever carbon
 117 remains stays in the part. The reference atmosphere is 4% H_2 in nitrogen
 118 humidified at a dew point of 20 °C, so that $x_{\text{ref}} = 0.023$ and $x_{\text{H}_2}^{\text{ref}} = 0.04$.
 119 The rate constant k_g is parameterised by a half-life $t_{1/2}$ at 800 °C in that
 120 atmosphere and an activation energy; no measured value exists for char on
 121 either material, so both enter the sensitivity analysis. The square of the
 122 Thiele modulus, $k_g L^2 / D_{\text{eff}} \sim 10^{-4}$ for $L = 2 \text{ mm}$, shows that gasification
 123 is kinetically controlled, so steam access is represented by the open-pore
 124 accessibility $a(\varepsilon_o)$ rather than by a diffusion equation.

125 Pyrolysis gas raises the pore pressure. The overpressure is computed
 126 from the compressible Darcy equation with Kozeny–Carman and Knudsen
 127 contributions to the permeability and a polymer-permeation term that carries

128 the flux before the pore network percolates; it is compared with the green or
 129 brown strength through a safety factor. The same slab tracks the through-
 130 thickness temperature difference and converts it into a thermal-stress index.

131 2.3. Densification

132 Both materials densify according to the Skorohod–Olevsky viscous sinter-
 133 ing model [48]. For a porous body of porosity θ and matrix shear viscosity η ,
 134 the stress is

$$\boldsymbol{\sigma} = 2\eta [\varphi(\theta) \dot{\boldsymbol{\epsilon}}' + \psi(\theta) \text{tr}(\dot{\boldsymbol{\epsilon}}) \mathbf{I}] + P_L(\theta) \mathbf{I}, \quad (3)$$

135 with $\varphi = (1 - \theta)^2$, $\psi = \frac{2}{3}(1 - \theta)^3/\theta$ and sintering stress $P_L = 3\gamma(1 - \theta)^2/r$,
 136 where γ is the surface energy and r half the Sauter mean particle diameter.
 137 A free body densifies at $\text{tr}(\dot{\boldsymbol{\epsilon}}) = -(P_L - \Delta p)/(2\eta\psi)$, where Δp is the pressure
 138 of insoluble gas trapped in closed pores. The two materials differ only in η , γ
 139 and r .

140 *Glass-ceramic.* The residual melt is Newtonian. Its viscosity follows the
 141 MYEGA equation [49],

$$\begin{aligned} \log_{10} \eta_g = \log_{10} \eta_\infty + (12 - \log_{10} \eta_\infty) \frac{T_g}{T} \\ \times \exp \left[\left(\frac{m}{12 - \log_{10} \eta_\infty} - 1 \right) \left(\frac{T_g}{T} - 1 \right) \right], \end{aligned} \quad (4)$$

142 with $\log_{10} \eta_\infty = -3$ (Pas), glass transition T_g (defined at 10^{12} Pas) and
 143 fragility index m . Crystallisation during firing follows non-isothermal Johnson–
 144 Mehl–Avrami–Kolmogorov kinetics [50] written in terms of the extended-
 145 volume integral,

$$\dot{\xi} = k_c(T) = k_{c,p} \exp \left[-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{T_{p,c}} \right) \right], \quad X = 1 - \exp(-\xi^n), \quad (5)$$

146 which avoids the singularity of the rate form at $X = 0$ for $n > 1$. The constant
 147 $k_{c,p}$ is fixed numerically so that the simulated crystallisation exotherm at
 148 10 K min^{-1} peaks exactly at the measured temperature $T_{p,c}$; Kissinger’s closed
 149 form [51] is exact only for $n = 1$ and would misplace the peak by about 1 K
 150 for $n = 2$. Crystals stiffen the melt as rigid inclusions,

$$\eta = \eta_g(T) \left(1 - \frac{X}{X_m}\right)^{-[\eta]X_m} \left(1 - \frac{f_C}{0.64}\right)^{-[\eta]0.64}, \quad (6)$$

151 the Krieger–Dougherty law [52] with intrinsic viscosity $[\eta] = 2.5$ and maximum
 152 packing X_m . The second factor treats residual char particles, volume fraction
 153 f_C , the same way. Once X approaches X_m the body stops flowing, which is
 154 how crystallisation arrests densification.

155 *Copper.* The copper skeleton creeps by diffusion. Its matrix viscosity combines
 156 lattice and grain-boundary diffusion [53],

$$\eta_{Cu} = f_\eta \frac{k_B T G^2}{42 \Omega (D_v + \pi \delta D_b / G)} \left(1 + \frac{C^2}{C_{inh}^2}\right) \left(1 - \frac{f}{0.64}\right)^{-1.6}, \quad (7)$$

157 where G is the grain size, Ω the atomic volume and f_η a calibration factor
 158 fitted to published copper densification data (Appendix A). The carbon term
 159 C (in ppm) describes a retarding effect of residual carbon on the sintering of
 160 copper. Its scale, $C_{inh} = 200$ ppm, is an estimate, and its influence on the
 161 results is examined in Section 4.10; and the last factor accounts for an inert
 162 ceramic filler of volume fraction f added to the conductor paste, a standard
 163 way to slow a metal paste down to the pace of its ceramic [39]. Grain growth
 164 is parabolic, driven by grain-boundary diffusion and pinned by porosity. The
 165 filler also lowers the conductivity; together with porosity,

$$\frac{\sigma_e}{\sigma_{e,0}} = (1 - \theta)^{3/2} \frac{1 - f}{1 + f/2}, \quad (8)$$

the second factor being the Maxwell Garnett result for insulating inclusions [54].

2.4. Atmosphere and the metallic state of copper

Steam removes char but oxidises copper once the water-to-hydrogen ratio exceeds the equilibrium of $\text{Cu}_2\text{O} + \text{H}_2 \rightleftharpoons 2\text{Cu} + \text{H}_2\text{O}$,

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} < K_{\text{Cu}}(T) = \exp\left(-\frac{\Delta G^\circ(T)}{RT}\right). \quad (9)$$

During the burnout segment the inlet hydrogen is therefore tied to the steam content, $x_{\text{H}_2} = x_{\text{H}_2\text{O}}/(s K_{\text{Cu}}(T_{\text{B}}))$, with a safety factor $s = 0.2$. The copper stays metallic by construction, and the steam is as oxidising towards carbon as the metal allows. This is the principle of the steam co-sintering developed for glass-ceramic/copper substrates [14, 6]; here it is applied in a model that says how long, at what temperature and in how much steam. Carbon gasification itself is thermodynamically unconstrained over the whole window (Section 4.2); its limits are kinetic.

2.5. Mechanics of co-firing

Porous viscosities. Eq. (3) describes an isotropic, linear viscous solid with shear viscosity $G_v = \eta\varphi$ and bulk viscosity $K_v = 2\eta\psi$, hence uniaxial viscosity $E_v = 9K_vG_v/(3K_v+G_v)$, viscous Poisson ratio $\nu_v = (3K_v-2G_v)/[2(3K_v+G_v)]$ and biaxial viscosity $B = E_v/(1 - \nu_v)$. The free linear sintering rate is $\dot{\epsilon}_{\text{f}} = -P_{\text{L}}/(6\eta\psi)$. The viscosities and the free rate follow from the slab histories of T , θ , G or ξ , and carbon. Shrinkage is reported as engineering linear shrinkage when single materials are compared, and as logarithmic strain in the co-firing histories, where the free-strain mismatch is the difference of the logarithmic strains of the two materials.

189 *Camber.* By the elastic–viscous correspondence principle, a bilayer of two
 190 linear viscous layers bends at a curvature rate given by Timoshenko’s bimetal
 191 solution [55], with elastic moduli replaced by biaxial viscosities and thermal
 192 strain by the difference in free sintering rates [56, 38, 41]:

$$\dot{\kappa} = \frac{6(1+\lambda)^2(\dot{\epsilon}_{f,2} - \dot{\epsilon}_{f,1})}{h[3(1+\lambda)^2 + (1+\lambda\mu)(\lambda^2 + 1/(\lambda\mu))]}, \quad (10)$$

193 with $\lambda = h_1/h_2$, $\zeta = B_1/B_2$ and $h = h_1 + h_2$. Layer 1 is copper and layer 2
 194 the glass-ceramic; positive curvature is concave towards the copper. Eq. (10)
 195 is integrated over the firing with the layer thicknesses updated by their own
 196 shrinkage. The reference stack is a 0.1 mm copper plane on 1 mm of glass-
 197 ceramic. Curvatures above about 100 m^{-1} lie outside the small-deflection
 198 range of the solution and are used only to compare designs.

199 *Embedded conductors.* A conductor line buried in the body cannot curl; its
 200 misfit is taken up by stress. We represent line and surrounding body as two
 201 Maxwell elements in series, each with a porous elastic modulus and a biaxial
 202 viscosity:

$$\dot{\epsilon}_{f,GC} - \dot{\epsilon}_{f,Cu} = \dot{\sigma} \left(\frac{1}{E_{Cu}} + \frac{1}{E_{GC}} \right) + \sigma \left(\frac{1}{B_{Cu}} + \frac{1}{B_{GC}} \right). \quad (11)$$

203 A compliant line follows the body and carries B_{Cu} times the misfit rate; a
 204 stiff line forces the body to flow around it. Tension in the copper ($\sigma > 0$)
 205 opens voids and breaks the line; compression in the copper means the body is
 206 shrinking onto a stiff inclusion and is itself in hoop tension, the classic origin
 207 of radial cracks around conductors [36, 37]. Two damage indices compare
 208 these stresses with the strength of the partner that fails at that moment,

$$\Pi_{Cu} = \frac{\max(\sigma, 0)}{\sigma_{s,Cu}(\rho_{Cu})}, \quad \Pi_{GC} = \frac{\max(-\sigma, 0)}{\sigma_{s,GC}(\rho_{GC})}, \quad (12)$$

209 where the strengths σ_s grow from the binder-carried green strength through
 210 the neck-controlled brown strength. The series coupling carries an Eshelby
 211 factor of order unity [57] that we do not resolve, so the indices are used
 212 comparatively. The line damage index reported below is the larger of the
 213 two, $\Pi_{\text{line}} = \max(\Pi_{\text{Cu}}, \Pi_{\text{GC}})$. The three-dimensional model below resolves the
 214 same coupling in full.

215 *Three-dimensional co-sintering.* The printed part is meshed voxel by voxel
 216 with trilinear hexahedra, so the mesh follows the layout of the printed part
 217 at the resolution of the voxel. Each element carries its material, porosity
 218 and microstructural state. The velocity field satisfies $\nabla \cdot \boldsymbol{\sigma} + \rho_b \mathbf{g} = \mathbf{0}$
 219 with Eq. (3), gravity, and a setter plate beneath the part: zero normal
 220 velocity and regularised Coulomb friction ($\mu_f = 0.6$) iterated to convergence
 221 at each step. Volumetric terms use one-point quadrature, which avoids
 222 locking as $\theta \rightarrow 0$. The problem is assembled with scikit-fem [58] and solved
 223 by conjugate gradients preconditioned with smoothed-aggregation algebraic
 224 multigrid carrying the six rigid-body modes [59]. Porosity is advanced with
 225 the exact mass-conserving update $1 - \theta^{n+1} = (1 - \theta^n) \exp(-\text{tr } \dot{\boldsymbol{\epsilon}} \Delta t)$. The
 226 temperature and carbon histories are taken from the two slab simulations.
 227 Below 0.3% porosity the constitutive law is evaluated at 0.3%, since the bulk
 228 viscosity otherwise grows without bound and the remaining densification no
 229 longer affects the shape. The simulation ends when the glass-ceramic reaches
 230 98% density or 90% crystallinity, whichever comes first; the remaining linear
 231 shrinkage of the body is then a few tenths of a per cent.

232 *Cooling.* Below a stress-free temperature T_{sf} , where the residual glass stops
 233 relaxing, the thermal-expansion mismatch is carried elastically by the glass-

ceramic and elastoplastically by the copper. The copper layer is treated as elastic-perfectly plastic with a temperature-dependent yield stress; its mean stress is capped at yield by a uniform plastic strain, and the curvature follows from Timoshenko’s elastic solution for the remaining elastic misfit.

2.6. Numerical solution, design search and sensitivity

The slab equations are integrated with a linearly implicit Rosenbrock method of orders 2(3) [60] on a geometrically graded mesh that is finest at the surface. A co-firing evaluation, two slab simulations and the mechanical post-processing, takes about 6 s. Designs were searched by differential evolution [61] over eleven variables: copper median particle size, filler fraction and solids loading; burnout temperature, time and steam fraction; temperature and duration of an intermediate matching hold; heating rate to the peak; peak temperature and hold. The objective was the largest free-strain mismatch between the two materials during firing, $\max_t |\varepsilon_{f,GC} - \varepsilon_{f,Cu}|$, which governs both camber and conductor stress. It was minimised subject to hard constraints and to a cycle-time budget of 18 h, the ε -constraint form of the trade-off between cycle time and mismatch. The constraints were: carbon at glass pore closure below 100 ppm and in the copper below 200 ppm; relative densities of at least 0.97 for the glass-ceramic and 0.92 for the copper; a crystallised fraction of at least 0.8; a line damage index below 1; no trapped-gas pressure above 1.2 times the sintering stress; and a positive margin to the melting point of copper. Violations entered the objective as a penalty proportional to their size, so the search is steered back towards feasible designs. Sobol initialisation rounds the population up to a power of two, so 128 designs were evolved for 15 generations after initialisation (2048 evaluations). The

259 objective still fell by 7% in the last generation, so the result is the best design
260 found rather than a proven optimum; it is called the nominal optimum below.

261 A second, robust search asked for a design that works across the uncertainty
262 of the two inputs that decide the timing, the gasification rate and the glass
263 transition. It evaluates each design at two corners of that uncertainty. At the
264 first, gasification is slow ($t_{1/2} = 4$ h) and the transition low ($T_g - 15$ K), so
265 the glass tends to seal before the char has gone. At the second, gasification is
266 fast ($t_{1/2} = 1$ h) and the transition high ($T_g + 15$ K), so the copper tends to
267 be released before the glass is ready. The objective is the larger of the two
268 peak mismatches; all constraints must hold at both corners, and the copper
269 must in addition reach 80% IACS. The cycle-time budget was relaxed to 20 h,
270 because the first feasible designs found needed about 20 h; the final design
271 needs 16 h, so the budget did not bind. The search was seeded with one such
272 feasible design, and a population of 128 was evolved for eight generations
273 (1152 evaluations, each over both corners). Only the two corners described
274 were used in the search; the other two corners were checked on the final
275 design. The nominal case lies between the corners and was checked on the
276 final design.

277 The influence of uncertain inputs was screened with Morris elementary
278 effects [62]: 16 trajectories through a four-level grid over fourteen factors,
279 240 co-firing evaluations in total, around the nominal optimum. Twelve
280 factors are the uncertain inputs with the ranges of Table 1; the other two are
281 manufacturing tolerances on the copper paste, -25 to $+33\%$ in particle size
282 and ± 0.05 in filler fraction. For each output the mean absolute elementary
283 effect μ^* measures the overall influence of a factor and the standard deviation

284 σ its non-linearity and interaction with other factors.

285 **3. Parameters and calibration**

286 *3.1. Provenance of the inputs*

287 No measurements were made for this study. Every input therefore comes
288 from one of five sources, and Table 1 states which one applies. A value is
289 labelled L when it was reported for the glass-ceramic of the IBM process or
290 for copper itself; A when it was taken from an analogous material, usually a
291 commercial copper-compatible LTCC; C when it was calibrated here against
292 published process observations (Section 3.2); D when it follows from other
293 entries of the table; and E when it is an engineering estimate. Most estimated
294 and analogue values that matter enter the sensitivity analysis of Section 4.10,
295 with the ranges given in the table; three that do not were checked one at a
296 time.

297 The reference glass is the MgO-rich cordierite-type glass used by IBM
298 for its copper-compatible substrates (glass #12: 52.5 SiO₂, 22.0 MgO, 22.0
299 Al₂O₃, 1.5 P₂O₅, 0.5 B₂O₃, 1.5 ZrO₂, wt%) [13, 63]. It was chosen because,
300 among copper-compatible glass-ceramics, it is the one for which the firing
301 process, the atmosphere and the reasons behind the chosen temperatures are
302 best documented. We could not obtain published viscosity data for it, so its
303 viscosity was calibrated.

304 *3.2. Calibration of the glass against the IBM process*

305 The patents that describe the IBM steam co-sintering process state, for
306 glass #12, why the burnout hold sits where it does. The hold is placed

Table 1: Model inputs. Basis: L, literature value for this material; A, analogue material; C, calibrated in Section 3.2; D, derived; E, estimate. Ranges in the last column are those used in the Morris screening.

Quantity	Symbol	Value	Basis	Source, remark or screening
<i>Glass-ceramic body</i>				
Median particle size	d_{50}	3.0 μm	A	IBM glass powders 2–7 μm
Size span $(d_{90} - d_{10})/d_{50}$	–	1.5	E	
Solids loading of the slurry	ϕ_{GC}	0.45	E	0.40–0.50
Density of the dense body	ρ_s	2600 kg m^{-3}	A	dense α -cordierite glass-ceramic [64]
Surface energy	γ	0.36 J m^{-2}	L	calculated for a cordierite-type glass, $\gamma = 0.36$, since water vapour low surface energy, sintering of silicate glass [66]
Glass transition (10^{12} Pa s)	T_g	734 $^{\circ}\text{C}$	C	714–754 $^{\circ}\text{C}$
Fragility index	m	32.4	C	28–40
High-temperature viscosity limit	$\log_{10} \eta_{\infty}$	–3	L	[49]
Crystallisation activation energy	E_c	303 kJ mol^{-1}	L	IBM-type high-cordierite glass-ceramic 440 kJ mol^{-1} in other MgO– Al_2O_3 – SiO_2 [68, 69]; 250–470 kJ mol^{-1} in other glass-ceramics [69]
Avrami exponent	n	2	E	1.5–3.7; 2.8–3.7 in a related glass [69]
Exotherm peak at 10 K min^{-1}	$T_{\text{p,c}}$	1040 $^{\circ}\text{C}$	C	1000–1080 $^{\circ}\text{C}$
Maximum crystal packing	X_m	0.64	E	random close packing
Density at pore closure	ρ_{cl}	0.92	E	checked at 0.88 and 0.95
Thermal conductivity, dense	k	2.8 $\text{W m}^{-1} \text{K}^{-1}$	A	copper co-fired LTCC GL570 [70]
Young’s modulus, dense	E_{GC}	128 GPa	A	GL570 [70]
Poisson’s ratio	ν_{GC}	0.25	E	
Thermal expansion coefficient	α_{GC}	3.0 ppm K^{-1}	A	production glass-ceramic 1.8 ppm K^{-1} for glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [70]
Flexural strength	$\sigma_{\text{s,GC}}$	210 MPa	L	glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [70]
Stress-free temperature	T_{sf}	690 $^{\circ}\text{C}$	D	strain point ($10^{13.5}$ Pa s) of glass #12 [70]

Copper conductor

Median particle size	d_{50}	1–20 μm	–	design variable; baseline 3 μm
Size span	–	1.2	E	
Solids loading	ϕ_{Cu}	0.40–0.60	–	design variable; baseline 0.5
Inert filler fraction	f	0–0.40	–	design variable; baseline 0

307 at 785 ± 10 °C, between the annealing and softening points of the glass.
 308 At 750 °C carbon removal would take “prohibitively excessive amounts of
 309 time”; at 830 °C the glass pores close and trap water and binder residue
 310 [13]. A later IBM patent on the co-sintering of multilayer ceramic substrates
 311 states that 33–50% of the total fired shrinkage occurs during the burnout
 312 step [78]; we assume that this holds for glass #12, and a third places the
 313 onset of crystallisation at about 900 °C, after the burnout hold and before
 314 the crystallisation hold at about 960 °C [79]. We simulated the full worked
 315 schedule of Ref. [13] (N_2 to 200 °C, $\text{H}_2/\text{H}_2\text{O}$ to 450 °C, 2.9 K min^{-1} to the
 316 hold, 6 h at $\text{H}_2/\text{H}_2\text{O} = 10^{-4}$, 0.5 h in N_2 , 2.1 K min^{-1} to 960 °C, 2 h) and
 317 used these statements as targets. The patent specifies the $\text{H}_2/\text{H}_2\text{O}$ ratio but
 318 not the dilution; we assumed 50% steam in nitrogen.

319 Two of them fix the viscosity. Requiring that a 6 h hold at 780 °C
 320 produce 40% of the fired shrinkage with the pores still open, and that the
 321 pores close within 2 h at 830 °C, gives $\log_{10} \eta = 10.62$ at 780 °C and 9.37 at
 322 830 °C. The MYEGA curve through these two points, with $\log_{10} \eta_\infty = -3$,
 323 has $T_g = 733.6$ °C and $m = 32.4$. The remaining observations were not used
 324 in the fit and serve as checks. Between 800 and 860 °C the calibrated viscosity
 325 falls by one decade per 43 K. Giess et al. found that the sintering of an IBM
 326 cordierite-type glass over the same range followed the viscosity of the glass,
 327 which fell by about one decade per 40 K [65]. The annealing point, taken
 328 as 10^{12} Pa s, lies at 734 °C and the Littleton softening point ($10^{6.6}$ Pa s) at
 329 980 °C, so the 785 °C hold lies between them as the patent requires; so does
 330 the dilatometric softening point, near 10^{10} Pa s, which the calibrated glass
 331 reaches at about 803 °C. Stoichiometric cordierite glass is more viscous by 3.7

Table 2: The IBM worked schedule [13] simulated at three hold temperatures. Shrinkage is the fraction of the total fired shrinkage reached by the end of the 6 h hold; carbon is the residual carbon in the glass-ceramic at the end of the hold.

Hold temperature	750 °C	785 °C	830 °C
Shrinkage during hold	7%	47%	100%
Pores at end of hold	open	open	closed
Carbon at end of hold (ppm)	33	0.006	0.07
Final relative density	0.990	0.990	0.990
Final crystallised fraction	1.00	1.00	1.00
Statement in [13]	too slow	chosen	pores close

332 decades at 780 °C, 1.0 at 900 °C and 0.4 at 960 °C [80]. Glass #12 contains
333 more MgO and less Al₂O₃ than cordierite, together with B₂O₃ and P₂O₅, so
334 a lower viscosity is plausible, but this comparison is not a test of the fit.

335 The crystallisation kinetics were treated in the same way. With the acti-
336 vation energy measured on an IBM-type glass ($E_c = 303 \text{ kJ mol}^{-1}$ [67]) and
337 $n = 2$, placing the onset of crystallisation (5% crystallised) at 900 °C in the
338 worked schedule requires an exotherm peak at 1040 °C at 10 K min^{-1} . The
339 simulated exotherm then spans 978–1066 °C, a width of about 90 K, compa-
340 rable to the 80–100 °C span of sintering temperatures that the IBM patent
341 finds covered by the crystallisation exotherm of its glasses [63]. Differential
342 thermal analysis of other MgO–Al₂O₃–SiO₂ glasses places the α -cordierite
343 peak at 1006–1032 °C [64, 69].

344 Table 2 summarises the result. The 785 °C hold removes the char and
345 leaves 47% of the shrinkage to the burnout step, inside the 33–50% reported
346 for the process, while the pores stay open. At 830 °C the body densifies

347 completely during the hold and seals its pores. These two rows reproduce the
348 calibration targets and are therefore not independent evidence. The 750 °C
349 row is independent: it depends on the gasification kinetics, which were not
350 fitted, and it leaves four orders of magnitude more carbon than the 785 °C
351 hold. The model does not reproduce the absolute time scale of the patent’s
352 statement. With the estimated gasification rate, 33 ppm of carbon remain
353 after 6 h at 750 °C, and whether that would have been acceptable cannot be
354 judged, because no carbon specification was published. Nor does the model
355 reproduce the reason the patent gives for avoiding 830 °C: the carbon has
356 gone before the pores close, and the trapped water that the patent mentions
357 is not modelled. For all three holds the final body is 99% dense and fully
358 crystallised, and its free linear shrinkage is 23.1%.

359 4. Results

360 4.1. Verification

361 The model was checked against closed-form solutions before it was used
362 for design (Appendix B). Isothermal densification of the glass-ceramic follows
363 the Skorohod–Olevsky solution to within 1.3×10^{-4} in porosity for 4 to 16
364 cells (Fig. 2a), and the three-dimensional curvature rate of a free copper/glass-
365 ceramic plate converges to the viscous Timoshenko value at least at second
366 order, with an error of 0.05% at 0.15 mm voxels (Fig. 2b). The analytical
367 camber of Eq. (10) and the finite-element model therefore describe the same
368 mechanics, and the analytical form can be used inside the design search.

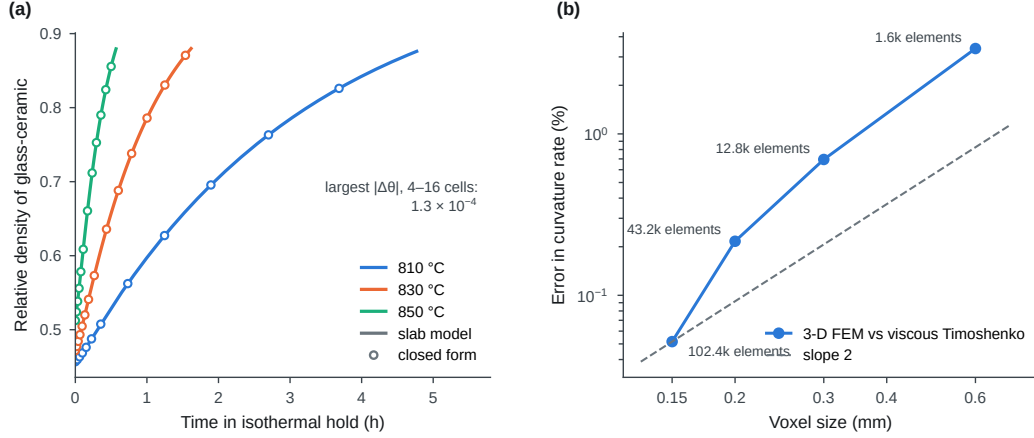


Figure 2: Verification. (a) Isothermal densification of the glass-ceramic slab at 810, 830 and 850 °C (lines, 8 cells) against the closed-form solution integrated along the simulated part temperature (circles); crystallisation and char were switched off. (b) Error of the three-dimensional curvature rate of a free 12×12 mm plate (0.6 mm copper on 1.8 mm glass-ceramic, 900 °C) against the viscous Timoshenko solution, Eq. (10), as the voxel size is refined.

4.2. Atmosphere window for burnout

Fig. 3 places the burnout atmosphere in the plane of temperature and $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$. Copper remains metallic below the orange line, which falls from $10^{6.2}$ at 500 °C to $10^{4.8}$ at 780 °C and $10^{4.1}$ at 1000 °C. Carbon gasification by steam is thermodynamically possible above the dashed line, which lies eight or more decades lower. The two constraints therefore leave a wide band of permissible atmospheres, and the choice within it is kinetic. At 780 °C, 30% steam with hydrogen at 20% of the copper limit ($p_{\text{H}_2\text{O}}/p_{\text{H}_2} = 1.2 \times 10^4$) gives a char half-life of 0.43 h. Forming gas (4% H_2) humidified through a bubbler at 20 °C, a common choice for copper, contains 2.3% water; its half-life at the same temperature is 3.1 h, seven times longer, because the steam partial

380 pressure is low and hydrogen inhibits the reaction.

381 The operating point of the IBM process, $H_2/H_2O = 10^{-4}$ at 785 ± 10 °C
382 [13], lies at 17% of the copper limit and almost coincides with the steam
383 point in Fig. 3. The rule used here, hydrogen tied to steam so that the
384 water-to-hydrogen ratio sits at a fixed fraction of the copper limit, is therefore
385 consistent with the atmosphere used for tape-cast substrates. It also says
386 how the ratio should change if the burnout temperature changes: by a factor
387 of about 2.5 per 100 K, following $K_{Cu}(T)$.

388 *4.3. Removing carbon before the glass seals*

389 The char has to leave the glass before its pores close, and the glass
390 carries far more of it than the copper does. After pyrolysis in nitrogen the
391 glass-ceramic contains 22 500 ppm carbon relative to its solids, the copper
392 5 300 ppm. The difference follows from the densities: both slurries hold a
393 similar volume of binder, but a unit volume of glass-ceramic solids weighs
394 a third as much as the same volume of copper. Carbon on the glass is also
395 the more harmful, since whatever is sealed in reacts with the water released
396 during crystallisation and bloats the part [13].

397 Fig. 4 maps the carbon left in the glass-ceramic at the moment its pores
398 close, as a function of burnout temperature and hold time. In steam (Fig. 4a)
399 the 100 ppm contour runs from 16 h at 710 °C to 1.1 h at 830 °C, and the
400 window closes abruptly at 850 °C, where the glass seals during the hold. The
401 IBM hold of 6 h at 785 °C falls in the region with less than 10 ppm. In
402 humidified forming gas (Fig. 4b) only the longest hold, 16 h at 790 °C, brings
403 the carbon down to 350 ppm; every other combination leaves more than
404 1000 ppm. Residual carbon also slows the densification of the glass. Acting

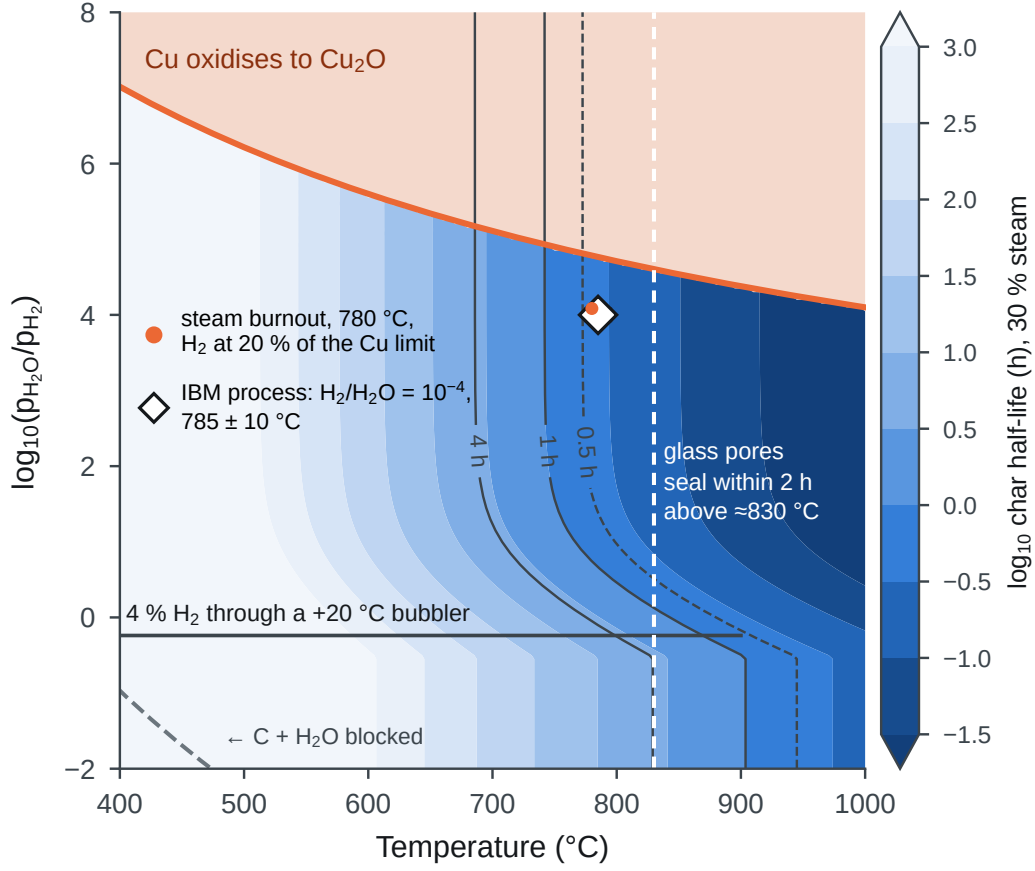


Figure 3: Burnout atmosphere window. Colour: half-life of char in 30% steam at the given $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$ (reference kinetics of Table 1). Copper oxidises above the orange line; carbon cannot be gasified to the left of the dashed line ($p_{\text{CO}} = 10^{-4}$ atm). Circle: steam burnout used in this work. Diamond: operating point of the IBM process [13]. The white line marks the temperature above which a 2 h hold seals the pores of the calibrated glass.

405 as rigid inclusions, the 0.03–2% of carbon retained in forming gas lowers the
 406 final density from 0.99 to 0.96–0.99. For this material pair, steam is therefore
 407 necessary rather than merely helpful.

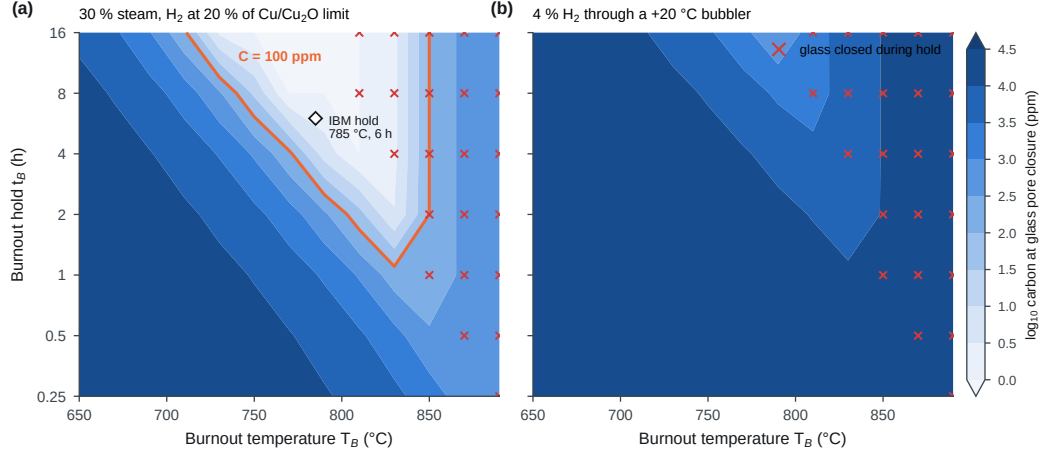


Figure 4: Carbon remaining in the glass-ceramic when its pores close, for burnout holds of temperature T_B and duration t_B within the baseline programme. (a) 30% steam with hydrogen at 20% of the Cu/Cu₂O limit; (b) 4% H₂ humidified at 20 °C. Orange: 100 ppm contour. Crosses: the glass sealed during the hold. Diamond: the IBM burnout hold [13].

4.4. Two densification ranges

Fig. 5 compares the free sintering of the two materials after binder removal at 5 K min⁻¹ in dry 4% H₂. The glass-ceramic shrinks by 23% between 850 and 933 °C (5 and 95% of its shrinkage) and then stops, because it crystallises: 93% of the glass has crystallised at 1060 °C. Fine copper densifies much earlier, and over a much wider range. A 3 µm powder shrinks between 454 and 702 °C; a 1 µm powder between 323 and 463 °C. Only for powders of 12 µm or coarser does the midpoint of densification fall inside the range of the glass, and these powders do not reach high density before 1060 °C: 83% for 12 µm and 60% for 20 µm.

The two curves also differ in shape. The glass densifies over a range of about 80 K, as a Newtonian melt whose viscosity falls steeply with tempera-

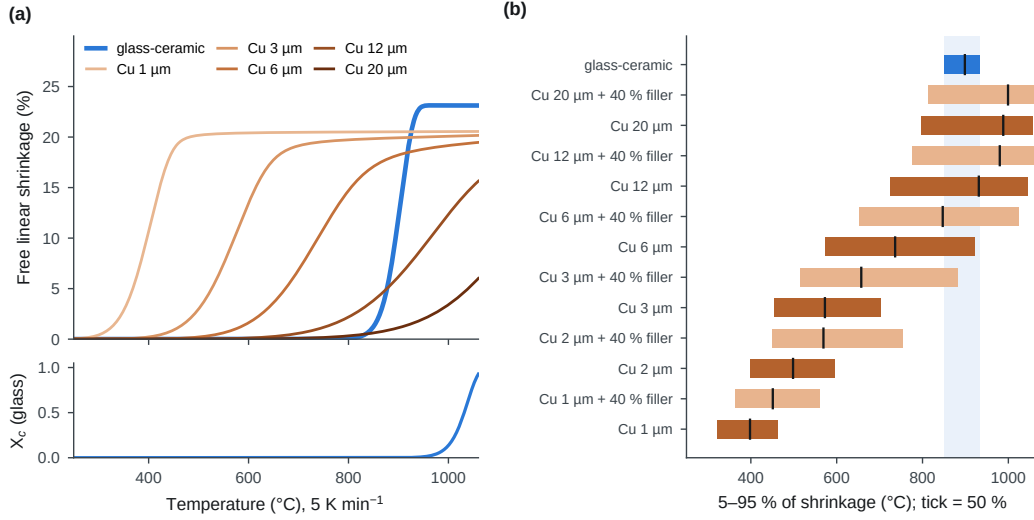


Figure 5: Free sintering of the two materials at 5 K min⁻¹ in dry 4% H₂ after binder removal. (a) Linear shrinkage of the glass-ceramic and of copper of different median particle size, and crystallised fraction of the glass. (b) Temperature range between 5 and 95% of the final shrinkage, with the midpoint marked, for copper with and without 40 vol% inert filler; the shaded band is the range of the glass-ceramic. Bars that reach 1060 °C had not finished densifying.

420 ture. Copper densifies over 140–350 K, because diffusional creep has a lower
 421 activation energy and grain growth slows it as it proceeds. Adding an inert
 422 filler to the paste delays and stretches the copper curve (Fig. 5b): 40% filler
 423 shifts the midpoint of 3 μm copper from 573 to 657 °C and widens the 5–95%
 424 range from 248 to 368 K. Neither particle size nor filler, on its own, gives
 425 copper a shrinkage curve that is as late and as steep as that of the glass. The
 426 matching has to come partly from the furnace programme.

4.5. *Co-firing with an unmatched paste*

The baseline design combines a 3 μm copper paste without filler with a programme that follows the IBM pattern: pyrolysis in nitrogen to 460 °C, a 4 h burnout at 780 °C in 30% steam, and sintering at 960 °C for 1 h in dry 4% H_2 (Fig. 6a). The design meets the chemical targets. The glass-ceramic retains 33 ppm of carbon when its pores close and reaches 99.1% density and full crystallinity; the copper is 97.6% dense, retains 81 ppm of carbon and conducts at 96% IACS.

It fails the mechanical ones. As soon as steam begins to remove the char, the copper starts to sinter, while the glass is still rigid. About an hour before the end of the burnout hold, the logarithmic shrinkage of the copper is 20.6% and that of the glass 4.9%. This is where the free-strain mismatch peaks, at 15.6%, and where a free bilayer of 0.1 mm copper on 1 mm glass-ceramic would reach a curvature of 58 m^{-1} , concave towards the copper. The damage index of an embedded line reaches $\Pi_{\text{Cu}} = 1.6$ at 680 °C, during the ramp to the burnout temperature, when the line is pulled in tension by the rigid body around it while its own necks are still weak. The glass then catches up, and its shrinkage overtakes that of the copper at 889 °C. The bilayer bends the other way and reaches -259 m^{-1} at 930 °C. The final logarithmic shrinkages differ by 4%: 26.3% for the glass-ceramic against 22.3% for the copper. For this material pair the difficulty lies less in the final shrinkage than in the order in which the two materials shrink.

4.6. *Copper paste design maps*

Fig. 7 shows how far the materials alone can repair the baseline, with the programme left unchanged. The upper row varies the copper paste. Coarser

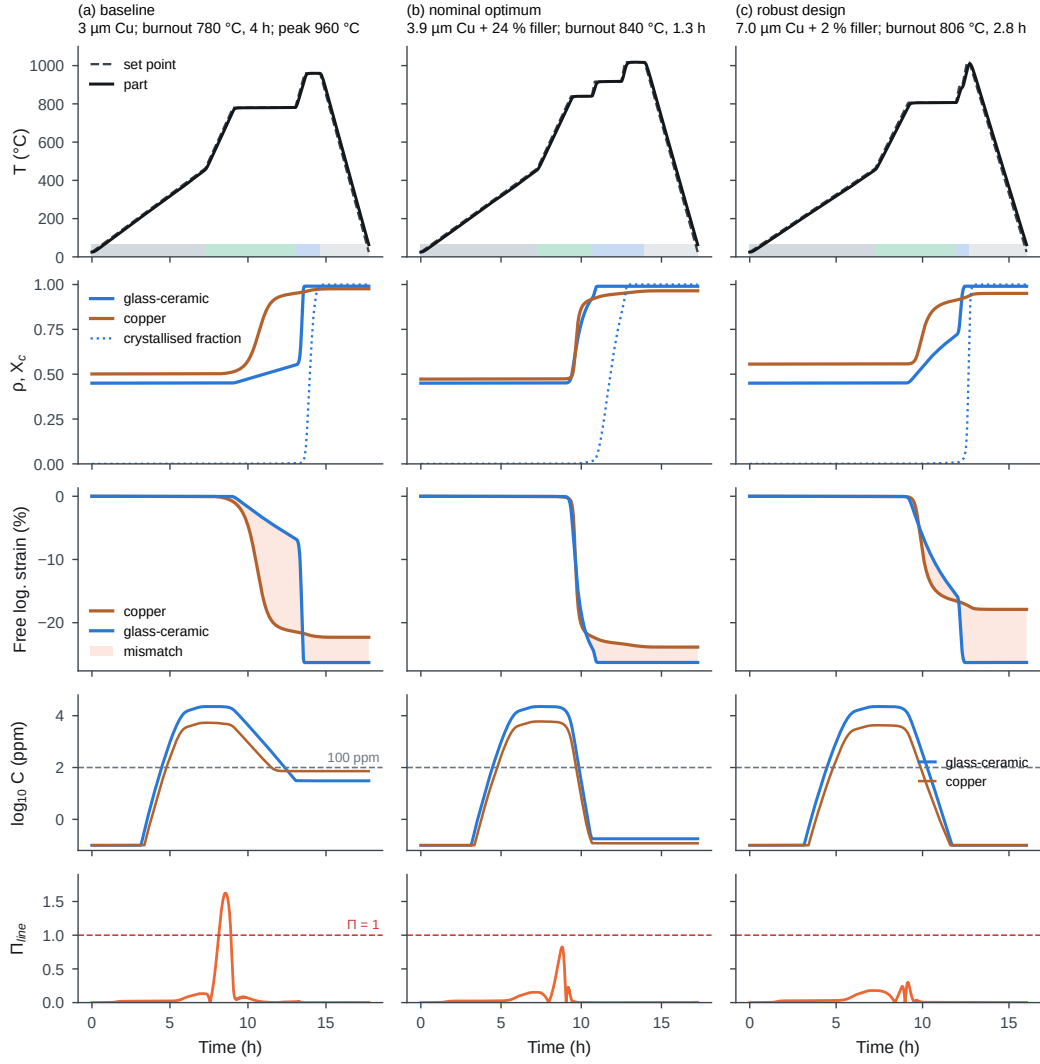


Figure 6: Co-firing histories of (a) the baseline, (b) the nominal optimum and (c) the robust design of Table 3. From top: furnace set point and part temperature, with the atmosphere of each segment shown by the band at the bottom (grey, N_2 ; green, steam; blue, dry H_2/N_2); relative densities and crystallised fraction of the glass; free logarithmic shrinkage of each material, the shaded area being the mismatch; carbon in the glass-ceramic and damage index of an embedded copper line.

powder and filler both delay the copper, and the damage index of an embedded
 line falls below 1 for particle sizes of 4.5 μm and above (Fig. 7b). The peak
 mismatch falls more slowly. Its lowest values, about 9.5%, are reached at
 6 μm with 30% filler and at 9 μm without filler; coarser pastes are released
 too late and the mismatch rises again, now with the copper behind the glass.
 Every step towards a later copper costs density or conductivity (Fig. 7c,d).
 Filler lowers the conductivity directly, by 14 points of IACS for every 10 vol%,
 while coarse powder lowers it through porosity: 18 μm copper reaches only
 76% density. Among the pastes that meet the density limit, 9 μm copper
 without filler gives the best compromise: 9.5% peak mismatch, a damage
 index of 0.38 and 91% IACS. None of them reduces the final camber, which
 stays between -190 and -260 m^{-1} for every paste that meets the density
 limit. The final camber is set by the difference in final shrinkage, which the
 paste barely changes.

The lower row varies the glass, with the baseline paste. The glass transition
 decides whether carbon is trapped. Below about 695 $^{\circ}\text{C}$ the glass seals during
 the 780 $^{\circ}\text{C}$ burnout and retains about 3000 ppm (Fig. 7e). Above that, the
 carbon stays below 100 ppm, but the higher the transition, the later the glass
 densifies, and the crystallisation peak has to rise with it if the glass is to reach
 full density before it crystallises (Fig. 7f). The crystallisation peak must not
 be too high either: above about 1080 $^{\circ}\text{C}$ the glass is less than 80% crystalline
 after 1 h at 960 $^{\circ}\text{C}$ (Fig. 7g). The most interesting region is the narrow band
 around $T_g = 700 \text{ }^{\circ}\text{C}$ (Fig. 7h). There the glass densifies during the burnout
 hold, together with the copper, and the peak mismatch falls to 6.5% while
 the carbon at closure stays near 50 ppm. A glass chosen to match the copper

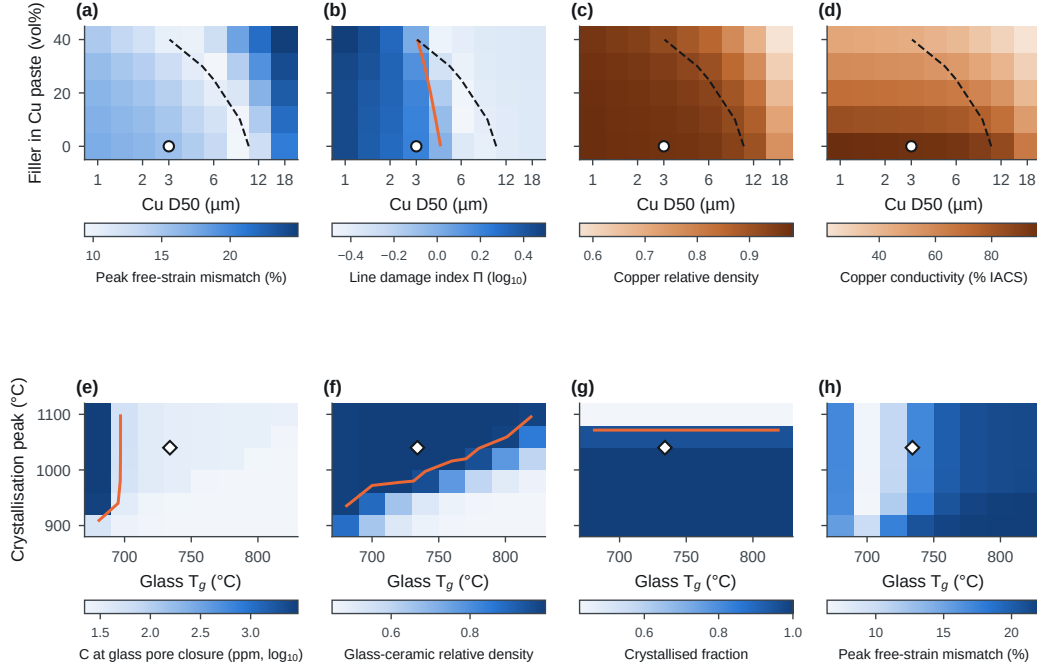


Figure 7: Design maps at the baseline programme. Top: copper paste, median particle size against filler fraction, with the baseline paste marked (circle): (a) peak free-strain mismatch; (b) damage index of an embedded line, orange contour $\Pi = 1$; (c) copper density; (d) conductivity. The dashed line is the density limit $\rho_{\text{Cu}} = 0.92$; pastes to its right fail it. Bottom: glass, transition temperature against crystallisation peak, with the calibrated glass marked (diamond) and the baseline paste: (e) carbon at pore closure, orange contour 100 ppm; (f) density of the glass-ceramic, orange contour 0.97; (g) crystallised fraction, orange contour 0.8; (h) peak mismatch.

477 would thus do more than any paste, but only within a window of about 20 K
 478 in T_g , and with little margin on carbon at its lower edge.

479 4.7. A matched paste and programme

480 Neither the paste nor the glass alone brings the peak mismatch below about
 481 6%, so the paste and the programme were searched together (Section 2.6).

482 The best design found within an 18 h cycle, called the nominal optimum
483 below, is listed in Table 3 and its history is shown in Fig. 6b. It uses a 3.9 μm
484 copper powder with 24 vol% filler. The burnout is short and hot: 1.3 h at
485 840 °C in 81% steam, just below the temperature at which the glass would
486 seal during the hold (Fig. 4a). A 1.7 h hold at 916 °C follows, and then 1.2 h
487 at 1018 °C. The peak mismatch falls from 15.6 to 3.7%, the damage index
488 of an embedded line from 1.6 to 0.83, and the final camber of the reference
489 bilayer from -254 to -76 m^{-1} . The glass-ceramic is 99.0% dense and fully
490 crystallised, with 0.2 ppm of carbon at pore closure.

491 The nominal optimum works through the retarding effect of carbon on
492 copper sintering that the model assumes (Eq. (7)). The copper cannot sinter
493 while it carries char (Section 4.5), so the burnout releases it. Placing the
494 burnout just below the coalescence of the glass, and making it fast with a
495 high steam fraction, releases the copper at the moment the glass begins to
496 flow. The two materials then densify together (Fig. 6b), and the filler slows
497 the copper just enough to follow the glass rather than lead it. The glass is
498 already 99% dense when the hold at 916 °C begins. The hold crystallises it,
499 while the copper continues to densify, and the peak at 1018 °C finishes the
500 copper.

501 The optimum has two weaknesses. The first is conductivity: 24% filler
502 limits the copper to 64% IACS. The second is robustness. The design is
503 sensitive to the timing on which it rests, and the lower part of Table 3 shows
504 what happens when uncertain inputs are varied one at a time. If the char
505 gasifies half as fast as assumed, the carbon at closure rises from 0.2 to 76 ppm
506 and the peak mismatch from 3.7 to 11.1%; at a quarter of the assumed rate

507 the carbon limit is exceeded by an order of magnitude. A glass transition
 508 15 K higher or lower than calibrated raises the peak mismatch to about 10%.
 509 So does any change in the carbon inhibition scale of copper sintering, C_{inh} ,
 510 by a factor of four or five either way, and at 1000 ppm the damage index of a
 511 buried line exceeds 1. A factor of two in the viscosity of copper changes the
 512 mismatch less (5.8–6.2%), but halving it also takes the damage index above 1.
 513 A design that relies on the timing of char removal is therefore only as good
 514 as the gasification kinetics, the viscosity of the glass and the effect of carbon
 515 on copper, and none of them has been measured for this material pair.

516 The robust search (Section 2.6) therefore asked for a design that meets
 517 every constraint at two corners of the uncertainty in gasification rate and
 518 glass transition and, in addition, keeps the copper above 80% IACS. The
 519 result, listed in the last column of Table 3 and shown in Fig. 6c, follows a
 520 different strategy. It uses a coarser 7 μm copper with almost no filler and a
 521 higher solids loading, and burns out for 2.8 h at 806 °C in 83% steam, some
 522 40 K below the temperature at which the glass would seal during the hold. It
 523 then heats quickly to a short peak at 1017 °C and needs 16.0 h in all. Both
 524 materials start to densify during the burnout hold, the copper once its char
 525 has gone and the glass because it is already soft. Being coarse, the copper
 526 runs only a few per cent ahead of the glass; the glass overtakes it during
 527 the heating to the peak. At nominal inputs the peak mismatch is 9.3%, the
 528 conductivity 89% IACS and the damage index 0.30. In both search corners
 529 the carbon stays below 50 ppm, the damage index below 0.43 and the peak
 530 mismatch at or below 11.4%; the two other corners, not used in the search,
 531 give 9.1 and 9.7% with 34 ppm of carbon at most. Halving the gasification

532 rate or shifting the glass transition by 15 K moves the peak mismatch only
 533 between 9.4 and 9.9%, and the design is equally insensitive to the carbon
 534 inhibition of copper and to the hydrogen inhibition of gasification (9.0–9.6%).
 535 Doubling the viscosity of copper is its weak point: the copper density falls
 536 to 0.912, just below the limit. The robustness is paid for in mismatch. The
 537 nominal value is two and a half times that of the nominal optimum, and
 538 the final camber, -187 m^{-1} , is larger. Most of the remaining mismatch is a
 539 difference in final shrinkage, 8.4%, because the paste now carries more solids
 540 than the glass slurry (56 against 45 vol%), which helps the copper reach
 541 density without filler. A gasification rate four times slower than assumed
 542 defeats both designs, which then leave 1295 and 940 ppm of carbon in the
 543 glass.

544 *4.8. The printed test vehicle*

545 The three designs were then fired in three dimensions as the test vehicle of
 546 Fig. 1a, a printed radio-frequency system-in-package. Its glass-ceramic body
 547 measures $10 \times 10 \times 2.5 \text{ mm}$ and has an open die cavity of $4 \times 4 \text{ mm}$, 0.75 mm
 548 deep. The copper comprises a $3 \times 3 \text{ mm}$ die-attach pad on the cavity floor,
 549 carried by nine thermal vias of 0.5 mm; a ground plane in the second layer,
 550 solid under the die and meshed elsewhere (0.5 mm lines at 1.25 mm pitch); a
 551 four-turn solenoid inductor wound about an axis parallel to the surface; and
 552 a buried stripline between two rows of via fences, led by a via to a bond pad
 553 on the upper face. Copper makes up 9.7% of the volume. The part is meshed
 554 with 15 232 voxels of 0.25 mm and rests on a setter. The voxel is coarser
 555 than the pixel of the printer, and the thinnest conductors are one voxel thick.
 556 Each run starts after binder removal and ends when the glass-ceramic is 98%

Table 3: Baseline and optimised designs. Lower part: the two optimised designs re-evaluated with one input changed from its nominal value: the char gasification half-life $t_{1/2}$ (2 h), the glass transition T_g (734 °C), the copper viscosity factor f_η (0.28), the carbon inhibition scale of copper sintering C_{inh} (200 ppm) and the hydrogen inhibition constant of gasification K_H (25). Bold: carbon above 100 ppm; [†]line damage index above 1; [‡]copper density below 0.92.

	Baseline	Nominal optimum	Robust design
<i>Paste and programme</i>			
Copper d_{50} (µm) / filler (vol%) / solids (vol%)	3.0 / 0 / 50	3.9 / 24 / 47	7.0 / 2 / 56
Burnout: T (°C) / time (h) / steam (%)	780 / 4.0 / 30	840 / 1.3 / 81	806 / 2.8 / 83
Matching hold: T (°C) / time (h)	–	916 / 1.7	889 / 0.18
Peak: T (°C) / time (h)	960 / 1.0	1018 / 1.2	1017 / 0.15
Cycle time (h)	17.7	17.3	16.0
<i>Outcome at nominal inputs</i>			
Carbon at glass pore closure (ppm)	33	0.2	0.03
Glass-ceramic density / crystallised fraction	0.991 / 1.00	0.990 / 1.00	0.990 / 1.00
Copper density / conductivity (% IACS)	0.976 / 96	0.965 / 64	0.950 / 89
Peak free-strain mismatch (%)	15.6	3.7	9.3
Line damage index $\max(\Pi_{Cu}, \Pi_{GC})$	1.63	0.83	0.30
Final camber of reference bilayer (m ⁻¹)	–254	–76	–187
<i>Robustness: carbon at closure (ppm) / peak mismatch (%)</i>			
$t_{1/2} = 4$ h	–	76 / 11.1	37 / 9.9
$t_{1/2} = 8$ h	–	1295 / 17.9	940 / 14.4 [‡]
$T_g - 15$ K	–	84 / 9.9	0.06 / 9.4
$T_g + 15$ K	–	0.06 / 9.5	0.03 / 9.8
$f_\eta \times 2$	–	0.2 / 5.8	0.03 / 11.4 [‡]
$f_\eta / 2$	–	0.2 / 6.2 [†]	0.03 / 8.1
$C_{inh} = 50$ ppm	–	0.2 / 9.6	0.03 / 9.5
$C_{inh} = 1000$ ppm	–	0.2 / 9.8 [†]	0.03 / 9.1
$K_H = 5$	–	22 / 8.6	8 / 9.6
$K_H = 100$	32	–	0.0 / 10.2

557 dense, at 903–917 °C (Fig. 8). The setter keeps the lower face flat, so the
558 distortion appears on the upper surfaces: the top frame and the cavity floor.
559 Fig. 8 shows it as the out-of-plane deviation from a uniformly shrunk part.

560 Two features of the copper layout shape the distortion. The ground plane
561 is stiff in its own plane once its copper has densified. The glass above it
562 cannot shrink laterally and densifies by shrinking more in thickness, so the
563 surface sinks over the plane and stays high along the rim, which carries no
564 plane. The vertical copper, which means the thermal vias, the via fences and
565 the via to the bond pad, acts the other way. Where it has densified before
566 the glass, it props the surface up. In the baseline both effects are strong,
567 because the copper is dense long before the glass flows. The part shrinks
568 by 15.6% in plane, against 23.1% for the free glass-ceramic. Its top frame
569 deviates by 516 μm from flat, with a trough around the cavity and ridges over
570 the via fences (Fig. 8d). The cavity floor varies by 274 μm , and the cavity
571 opening shrinks by 12.8% while the body shrinks by 15.6%, so the fired cavity
572 is oversize relative to the rest of the part. In the nominal optimum the copper
573 and the glass densify together, and these effects almost vanish. The in-plane
574 shrinkage is 21.5%, the top frame is flat to 114 μm , the cavity floor to 62 μm ,
575 and the cavity opening follows the body to within 1%. The robust design lies
576 between them: 18.8% in-plane shrinkage, 266 μm on the frame and 167 μm
577 on the cavity floor.

578 The three designs rank in the same order by distortion and by the peak
579 mismatch of the one-dimensional analysis, which supports the use of that
580 analysis for design. The relief follows the peak mismatch rather than the final
581 one: the robust design ends with a larger final mismatch than the baseline

582 (8.4 against 4.0%) but with half its distortion. The absolute values depend
 583 on the layout and should be read with the coarse voxel in mind. The meshed
 584 plane and the via fences are standard features of co-fired packages, however,
 585 and the model shows that the order in which the copper and the glass densify
 586 decides how much they distort the part.

587 *4.9. Cooling from the stress-free temperature*

588 Below about 690 °C, the strain point of the calibrated glass, mismatch
 589 stresses no longer relax. The difference in expansion between copper (16.5–
 590 22 ppm K⁻¹ over this range) and the glass-ceramic (3 ppm K⁻¹) must then
 591 be carried by the solids. Fig. 9 shows the consequences for a copper plane
 592 on 1 mm of glass-ceramic. If copper remained elastic, its mean stress would
 593 exceed 900 MPa at room temperature and a 20 mm stack with 100 µm of
 594 copper would bow by 250 µm. Annealed copper cannot carry that stress. It
 595 yields within about 13 K of cooling and then follows its yield stress, which rises
 596 from about 27 MPa at 690 °C to 69 MPa at room temperature. The bow falls
 597 accordingly, to 13 µm for 100 µm of copper and to 6 µm for 50 µm, and the
 598 glass-ceramic stays in mild compression (–7 MPa for 100 µm of copper). The
 599 result does not depend on the stress-free temperature: starting the analysis
 600 at 500, 600, 700 or 800 °C gives the same final stress and bow, because the
 601 copper yields almost at once in every case. Plastic accommodation makes
 602 the thermal-expansion mismatch a secondary problem compared with the
 603 sintering mismatch, provided the copper is dense and ductile. The price is
 604 cyclic plasticity in the conductor, which matters for thermal cycling in service
 605 but is outside the scope of this study.

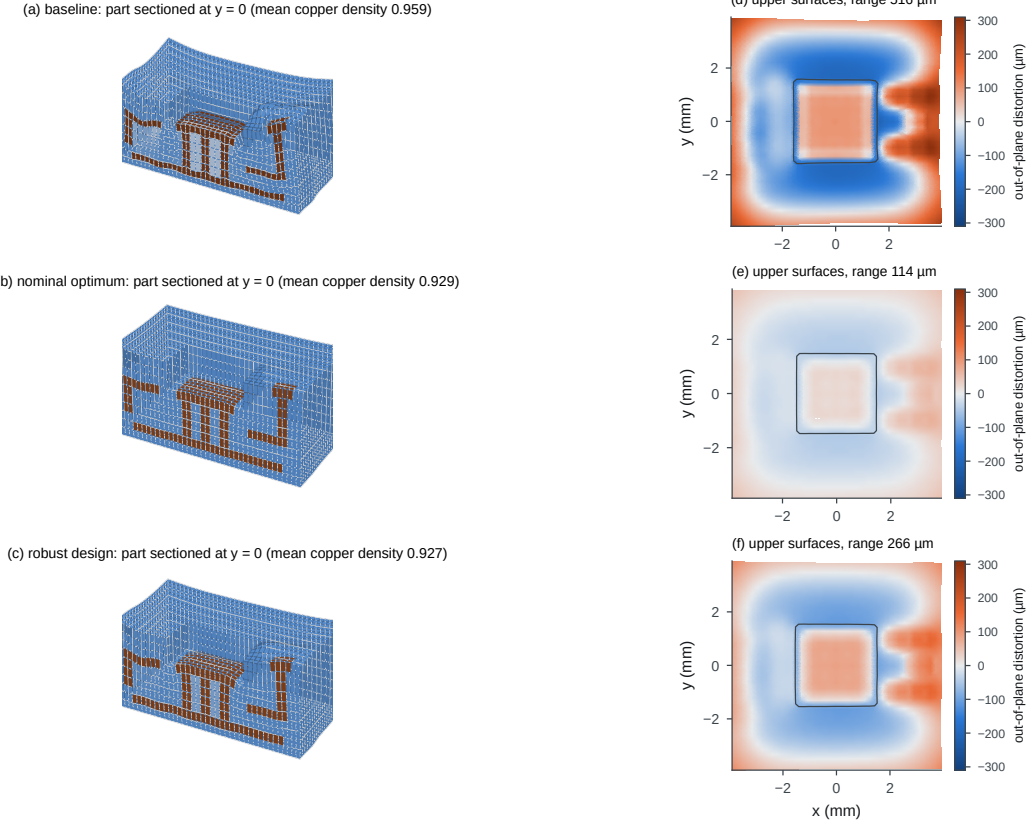


Figure 8: Three-dimensional co-sintering of the printed system-in-package (0.25 mm voxels, setter support, gravity) for the three designs of Table 3, from binder removal to 98% density of the glass-ceramic. Left (a–c): part sectioned at $y = 0$, through the solenoid, the die cavity with its pad and thermal vias, the ground plane and the stripline with its via to the bond pad; the section face is shaded darker. Right (d–f): out-of-plane distortion of the upper surfaces, the top frame and the cavity floor (outlined), relative to a uniformly shrunk part, on a common colour scale. The lower face is held flat by the setter.

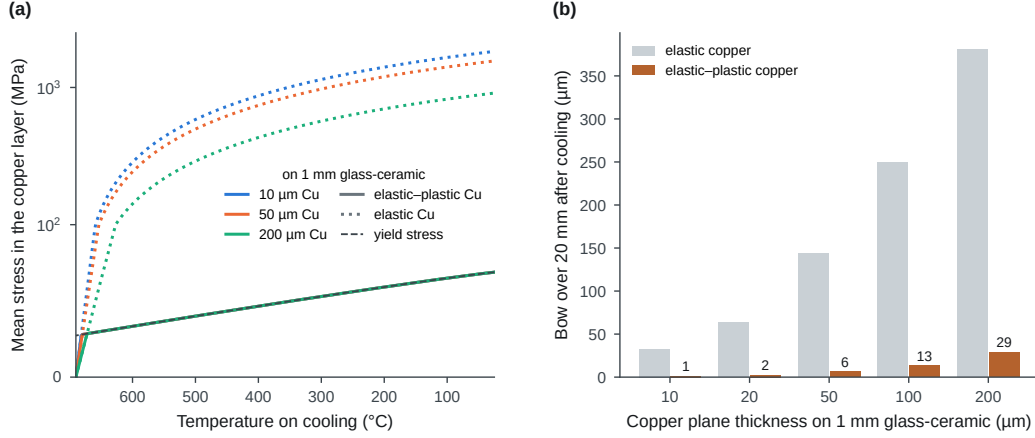


Figure 9: Cooling of a copper plane on 1 mm of glass-ceramic from the stress-free temperature (690 °C). (a) Mean stress in the copper layer for elastic and elastic-perfectly plastic copper; the plastic curves coincide with the temperature-dependent yield stress. (b) Bow over a 20 mm span at room temperature.

606 4.10. Which inputs decide the outcome

607 Fig. 10 ranks the fourteen uncertain inputs by their elementary effects
608 around the nominal optimum. Elementary effects are expressed per full
609 range of each factor. The half-life of char gasification dominates: across
610 its range of 0.5–8 h it changes the peak mismatch by 16 percentage points
611 on average and the carbon at glass closure by several orders of magnitude,
612 from well below 1 ppm to above the 100 ppm limit. It also comes second
613 for copper density, because char that is not removed keeps the copper from
614 sintering. The carbon trapped in the glass depends next on the fragility and
615 transition temperature of the glass, an order of magnitude less strongly. The
616 density of the glass-ceramic is governed by its own properties: the transition
617 temperature, the particle size and the crystallisation kinetics. The density
618 and conductivity of the copper depend on the viscosity factor f_η and on the

619 filler. The peak mismatch, the quantity that the design sought to minimise,
 620 responds to almost every input after the gasification rate: f_η , the glass
 621 transition, the char yield, and the particle size and solids loading of the glass.
 622 For every output that matters, σ is of the same order as μ^* . The effects are
 623 therefore strongly non-linear and interact, which is what a design based on
 624 the coincidence of two processes should be expected to show. The tolerances
 625 on the copper paste itself (particle size -25 to $+33\%$, filler ± 5 vol%) matter
 626 for the damage index of a buried line, together with f_η and the char yield,
 627 and the filler tolerance also for the conductivity. The Avrami exponent and
 628 the activation energy of crystallisation matter only for the density of the
 629 glass-ceramic, and the crystallised fraction (not shown) remained complete
 630 in every run. Three inputs were left out of the screening and checked one
 631 at a time instead (Table 3): the carbon inhibition scale of copper sintering,
 632 the hydrogen inhibition of gasification and the density at which the glass
 633 closes its pores. The last has little effect; the peak mismatch of the nominal
 634 optimum stays between 3.5 and 3.9% . The first two matter as much as the
 635 gasification rate for the nominal optimum, and hardly at all for the robust
 636 design.

637 5. Discussion

638 5.1. *The order of events in co-firing*

639 Taken together, the results fix an order of events that any firing programme
 640 for this material pair has to respect. Pyrolysis in nitrogen leaves about 2%
 641 of char, relative to the solids, in the glass-ceramic and about 0.5% in the
 642 copper. The char must then be gasified before the glass seals its pores. With

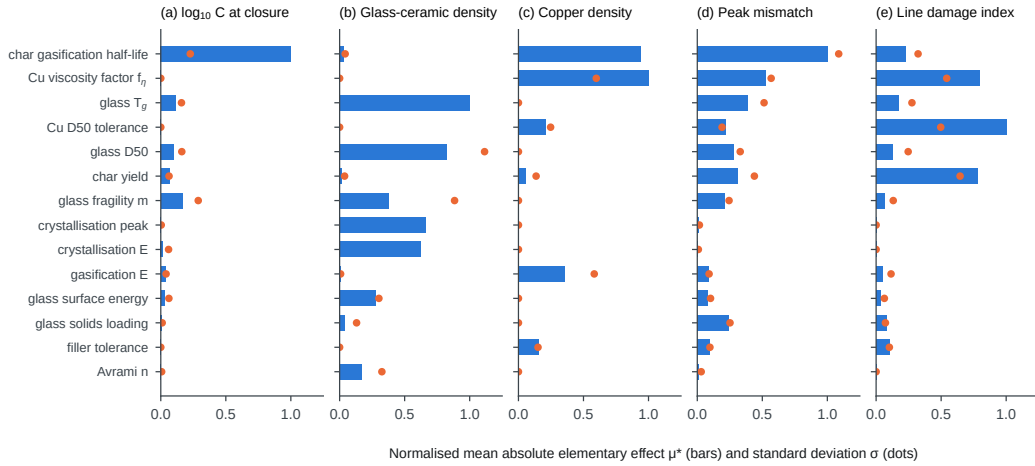


Figure 10: Morris screening of the fourteen uncertain inputs around the nominal optimum (16 trajectories, 240 co-firing evaluations). Bars: mean absolute elementary effect μ^* ; dots: standard deviation σ of the elementary effects, both normalised by the largest μ^* of each output. Factors are ordered by their summed normalised importance. Ranges as in Table 1; the copper particle size and filler fraction are varied as manufacturing tolerances around the design.

643 the atmosphere held at a fixed fraction of the copper–oxide equilibrium, this
 644 requires hours at 770–800 °C and is the slowest step of the process. In the
 645 model, removing the char also releases the copper. Residual carbon is assumed
 646 to retard the sintering of copper (Eq. (7)), so a fine powder that has been
 647 held back by its own char starts to densify as soon as the steam clears it, at
 648 680–780 °C in the baseline and well before the glass. This effect has not been
 649 measured for photopolymer char, and its strength, C_{inh} , is an estimate. In
 650 silver-metallised LTCC fired in air this coupling does not arise, because the
 651 binder burns off completely at low temperature and the metal is free to sinter
 652 from then on. Here the burnout step is also, whether intended or not, the
 653 first sintering step of the conductor. The glass then densifies over a range

654 of about 80 K and stops when it crystallises, and below the strain point of
655 the residual glass the expansion mismatch is taken up by plastic flow of the
656 copper.

657 This order explains why neither lever works alone. The furnace programme
658 cannot delay the densification of fine copper beyond the burnout, because the
659 burnout has to happen below the coalescence of the glass. The paste cannot
660 be made as late and as steep as the glass (Fig. 5) without losing conductor
661 density. The two designs of Table 3 take different approaches. The nominal
662 optimum uses the assumed coupling between char and copper to start both
663 materials together. It reaches the lowest mismatch, but it depends on the
664 timing of char removal and on the strength of the coupling, which are among
665 the least known inputs of the model. The robust design gives up part of that
666 gain and relies on a coarse copper that is intrinsically slow. It tolerates a
667 factor of two in the gasification rate together with 15 K in the glass transition,
668 and it is insensitive to the assumed effect of carbon on copper. Which of the
669 two is preferable depends on what has been measured. Once the gasification
670 kinetics, the viscosity of the glass and the effect of carbon on copper are
671 known much more tightly than these bounds, a design closer to the nominal
672 optimum becomes safe.

673 *5.2. Relation to tape-based co-firing*

674 The rule used for the atmosphere, hydrogen tied to steam so that the
675 water-to-hydrogen ratio sits at a fixed fraction of the copper limit, is consistent
676 with the operating point of the IBM process (Fig. 3), and the calibrated glass
677 is consistent with the burnout temperature chosen for it (Table 2). The model
678 adds no new principle to that process, but it makes it quantitative: how long

679 the burnout must last at a given temperature and steam content, how much
680 carbon is left when the pores close, and how the choice of copper powder
681 interacts with the programme. In LTCC, camber is controlled by adjusting
682 the paste formulation [39] or, with master-sintering-curve predictions, the
683 heating schedule [40]. For copper in a crystallising glass, Fig. 5 shows why
684 paste formulation alone is not sufficient. The heating programme sets the
685 order in which the materials are released to sinter, and the paste then adjusts
686 the rate.

687 Printing changes the problem in three respects. First, a photocurable
688 slurry carries more binder than a cast tape: at 45 vol% solids the glass-ceramic
689 shrinks by 23% linearly, against 12.7–15.8% in the plane of commercial LTCC
690 tapes [81, 82], and it holds correspondingly more char. Second, conductors
691 are no longer confined to printed planes connected by punched vias. The test
692 vehicle combines a die cavity, thermal vias, a meshed ground plane, a solenoid
693 wound about an axis parallel to the surface and a shielded stripline in one build.
694 Tape lamination would need stacks of punched vias for the solenoid alone.
695 In the three-dimensional model, the meshed plane and the vertical copper
696 both leave their imprint on the fired surfaces. A printed layout can shape
697 them freely, for example by grading the mesh or by balancing the via density
698 across the part. Third, printed ceramics may shrink anisotropically between
699 the build direction and the plane, as laminated tapes do: unconstrained IBM
700 laminates fired in H_2/H_2O shrink about half as much through the thickness
701 as in the plane [79]. The model treats both materials as isotropic. Anisotropy
702 would change the camber and the shape of the fired part, but not the order
703 of events described above.

704 5.3. *Limitations*

705 The results rest on published data and a model; no part has been fired.
706 Four inputs deserve particular caution. The viscosity of the glass was cali-
707 brated against statements in patents, not measured, and it describes one glass
708 composition; a different cordierite-type glass will shift the whole window. The
709 gasification kinetics of photopolymer char on either material are unknown
710 and were estimated, so the absolute times of the burnout carry the largest
711 uncertainty of the study (Section 4.10). The retarding effect of carbon on
712 copper sintering is an assumption of the model whose strength was estimated.
713 The viscosity factor of copper was fitted to a coarser powder than those
714 considered here. In addition, the model omits several phenomena that can
715 matter in practice: dissolution of copper oxide in the glass and interdiffusion
716 at the interface, adhesion and delamination, the water released by the glass
717 during crystallisation, the effect of residual carbon on dielectric loss, and
718 cyclic plasticity of the copper in service. The damage indices compare stresses
719 with strength estimates and are meant for ranking designs, not for predicting
720 failure. The three-dimensional runs stop at 98% density of the glass-ceramic,
721 so the final densification of the copper at the peak temperature, inside an
722 already rigid body, is represented only in the slab model.

723 5.4. *What to measure first*

724 The sensitivity analysis (Fig. 10) and the one-at-a-time checks of Table 3
725 turn the limitations into a short list of measurements, ordered by their value
726 to the design:

- 727 1. The gasification kinetics of the photopolymer char on each material, in
728 the steam–hydrogen atmosphere of the burnout. Interrupted burnout runs

729 followed by carbon analysis at three temperatures would fix both the rate
730 and its activation energy. No other quantity has as large an effect on the
731 carbon trapped in the glass and on the mismatch.

- 732 2. The viscosity of the glass between 750 and 900 °C, which fixes T_g and
733 m . Beam bending or parallel-plate viscometry of the glass would give it
734 directly, and so would isothermal dilatometry of the printed glass-ceramic,
735 interpreted with the closed form of Appendix B.
- 736 3. The sintering viscosity of the copper slurry, f_η , from dilatometry of printed
737 copper coupons in dry hydrogen after binder removal.
- 738 4. The char yield of both cured slurries, from thermogravimetry in nitrogen.
- 739 5. Whether, and how strongly, residual char slows the sintering of the copper
740 slurry, from dilatometry of copper coupons after partial and complete
741 burnout. The nominal optimum depends on it; the robust design does not.

742 The particle size and solids loading of the glass also matter, but they are
743 specified by the supplier and the slurry formulation. The crystallisation
744 kinetics matter little within the ranges considered, as long as the exotherm
745 peak lies above the sintering range. All of these measurements are routine,
746 and most of them can be made on single-material coupons printed with the
747 same slurries.

748 *5.5. Implications for three-dimensional packages*

749 A fired body of this kind would have the properties that made glass-
750 ceramic/copper attractive for high-end packaging. Its thermal expansion,
751 3.0 ppm K⁻¹, is close to that of silicon (2.6 ppm K⁻¹) [71], its permittivity is
752 about 5 [13], and its conductors are copper. From porosity and filler alone
753 the model predicts 89% IACS for the conductor of the robust design and 96%

754 for a paste without filler. IBM reported $2.0 \pm 0.2 \mu\Omega \text{ cm}$, about 86% IACS,
755 for copper co-fired in $\text{H}_2/\text{H}_2\text{O}$ [13]. The difference points to impurities and
756 grain-boundary oxygen, which the model does not include. Printing adds the
757 geometric freedom that tape lamination lacks. Vertical coils, shielded lines,
758 cavities and ground planes of arbitrary shape can be built into one part. That
759 freedom is relevant to antenna-in-package modules and millimetre-wave front
760 ends [3, 2], and to substrates for wide-bandgap power devices [4]. The present
761 study defines a candidate firing route for such parts and the few quantities
762 that have to be measured before it is tried.

763 6. Conclusions

764 A model of binder removal, sintering and crystallisation was used to design
765 the co-firing of copper inside a cordierite-type glass-ceramic printed as one
766 part by multi-material vat photopolymerisation. It was verified against closed-
767 form solutions, and its glass was calibrated against the published IBM glass-
768 ceramic/copper process, with whose atmosphere and burnout temperature
769 it is consistent. No experiments were made. The main conclusions are as
770 follows.

- 771 1. Of the atmospheres examined, only steam removes the carbon from the
772 glass before its pores close. Hydrogen tied to the steam at a fixed fraction
773 of the $\text{Cu}/\text{Cu}_2\text{O}$ equilibrium keeps the copper metallic and is consistent
774 with the IBM operating point. At 780°C such an atmosphere gasifies char
775 seven times faster than humidified forming gas, which leaves more than
776 300 ppm of carbon in the glass-ceramic in every programme examined.
- 777 2. Fine copper and the glass densify in different temperature ranges: 450–

778 700 °C for 3 µm copper against 850–930 °C for the glass. Co-fired with
779 the IBM-type programme, the copper densifies during the burnout, the
780 free-strain mismatch reaches 15.6%, and the damage index of a buried line
781 exceeds 1. The final shrinkages differ by only 4%; the difficulty lies in the
782 order in which the two materials shrink.

783 3. If char holds copper back, as the model assumes, the burnout step also
784 releases the copper to sinter. The best paste and programme found exploit
785 this coupling: a short, hot burnout just below the coalescence of the glass
786 releases both materials together and reduces the peak mismatch to 3.7%.
787 The design is sensitive to this timing. Halving the gasification rate, shifting
788 the glass transition by 15 K or changing the strength of the carbon effect
789 by a factor of four to five restores one half to two thirds of the mismatch
790 that the optimisation removed, and the 24% filler limits the copper to 64%
791 IACS.

792 4. A design required to meet every constraint under a factor of two either
793 way in the gasification rate combined with ± 15 K in the glass transition
794 uses 7 µm copper with almost no filler and a 2.8 h burnout at 806 °C. At
795 all four corners of that uncertainty it keeps the peak mismatch between
796 9.1 and 11.4%, the copper at 89% IACS and the carbon at glass closure
797 below 50 ppm, in a 16 h cycle, and it is insensitive to the assumed effect
798 of carbon on copper. The robustness costs a factor of two and a half in
799 nominal mismatch.

800 5. In a printed system-in-package with a die cavity, thermal vias, a meshed
801 ground plane, a solenoid and a shielded stripline, copper that densifies
802 before the glass restrains the body. The ground plane makes the glass

803 above it sink, and the vias prop the surface up. The baseline part shrinks
804 by 16% in plane instead of 23%, its top frame deviates by 0.52 mm from flat,
805 and its cavity ends oversize. The nominal optimum reduces the deviation
806 to 0.11 mm and the robust design to 0.27 mm, the same ranking as the
807 one-dimensional peak mismatch.

808 6. On cooling, plastic flow of the copper limits the stress it transmits to its
809 yield stress. A 100 μm copper plane on 1 mm of glass-ceramic bows by
810 13 μm over 20 mm, about a nineteenth of the elastic prediction.

811 7. The inputs that decide the outcome are, in order, the gasification kinetics
812 of the char, the viscosity of the glass, the sintering viscosity of the copper
813 and the char yield. All four can be measured on single-material printed
814 coupons, and they should be measured before the process is tried.

815 **CRedit authorship contribution statement**

816 **[Author names]:** Conceptualization, Methodology, Software, Formal
817 analysis, Visualization, Writing – original draft, Writing – review & editing.

818 **Declaration of competing interest**

819 The authors declare that they have no known competing financial interests
820 or personal relationships that could have appeared to influence the work
821 reported in this paper.

822 **Data availability**

823 The simulation code, input parameters with their provenance, and the
824 scripts that generate every figure in this paper are available from the corre-

825 sponding author on reasonable request.

826 **Declaration of generative AI and AI-assisted technologies in the** 827 **writing process**

828 During the preparation of this work the authors used [name of tool] in
829 order to [reason, e.g. assist with code development, data analysis and drafting
830 of the text]. After using this tool, the authors reviewed and edited the content
831 as needed and take full responsibility for the content of the publication.

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1137 **Appendix A. Copper slab model and its calibration**

1138 The copper conductor uses the slab model developed for single-material
1139 copper parts printed from a photocurable slurry. Besides the binder chemistry

1140 of Section 2.2, it tracks the oxygen bound in the powder as Cu_2O . The oxide
 1141 grows in oxygen-bearing gas by a decelerating law fitted to powder oxidation
 1142 data, is reduced by hydrogen above the equilibrium ratio of Eq. (9), and
 1143 reacts with char by $\text{Cu}_2\text{O} + \text{C} \rightarrow 2\text{Cu} + \text{CO}$. None of these oxidation paths is
 1144 active in the co-firing programmes studied here, because the inlet gas contains
 1145 no oxygen and the water-to-hydrogen ratio is held below the copper–oxide
 1146 equilibrium. What matters for co-firing is the sintering viscosity. Eq. (7) uses
 1147 the lattice and grain-boundary diffusion data for copper tabulated by Frost
 1148 and Ashby [53], with one factor, f_η , fitted to the densification of copper parts
 1149 printed from a 60 vol% photocurable slurry of 22 μm powder and sintered
 1150 for 4 h at 1050 $^\circ\text{C}$ in hydrogen, which reached 90–94% relative density [32].
 1151 The fit gives $f_\eta = 0.28$, with 0.22–0.34 spanning the reported density range.
 1152 The powder size enters Eq. (7) through the grain size, set initially to 0.4
 1153 times the median particle size, and through the sintering stress. Transferring
 1154 f_η from 22 μm to the 3–12 μm powders considered here therefore rests on
 1155 the diffusional scaling of Eq. (7); this is why f_η is included in the sensitivity
 1156 analysis of Section 4.10.

1157 **Appendix B. Verification**

1158 *Viscous densification of the glass-ceramic.* With crystallisation and char
 1159 removed, and open porosity, Eq. (3) with a Newtonian matrix gives $\dot{\theta} =$
 1160 $-9\gamma\theta/(4r\eta)$. The slab model was run through isothermal holds at 810, 830
 1161 and 850 $^\circ\text{C}$ with 4, 8 and 16 cells. The simulated porosity follows the closed
 1162 form, integrated along the simulated part temperature, to within 1.3×10^{-4}
 1163 over the full range of densification before pore closure, independently of the

1164 mesh (Fig. 2a).

1165 *Bilayer curvature.* The three-dimensional model of a free 12×12 mm plate
1166 made of 0.6 mm copper on 1.8 mm glass-ceramic was solved for the instan-
1167 taneous velocity field at 900 °C, and the curvature rate was fitted to the
1168 mid-plane deflection. It converges to the viscous Timoshenko value of Eq. (10)
1169 at least at second order: the error falls from 3.4% at 0.6 mm voxels to 0.69,
1170 0.22 and 0.05% at 0.3, 0.2 and 0.15 mm (Fig. 2b).

1171 *Free sintering and mass conservation.* A free, frictionless cube held isother-
1172 mally densifies homogeneously and isotropically, matches the closed-form
1173 Skorohod–Olevsky solution to 0.22% in density, and conserves solid mass to
1174 0.07%.

1175 *Crystallisation kinetics.* With the rate constant calibrated as described in
1176 Section 2.3, the simulated exotherm at 10 K min^{-1} peaks within 0.01 K of
1177 the prescribed temperature for Avrami exponents of 1, 2 and 3.